

Statistical Foundation for RNA-Seq Analysis

Instructor: Ryan Huang
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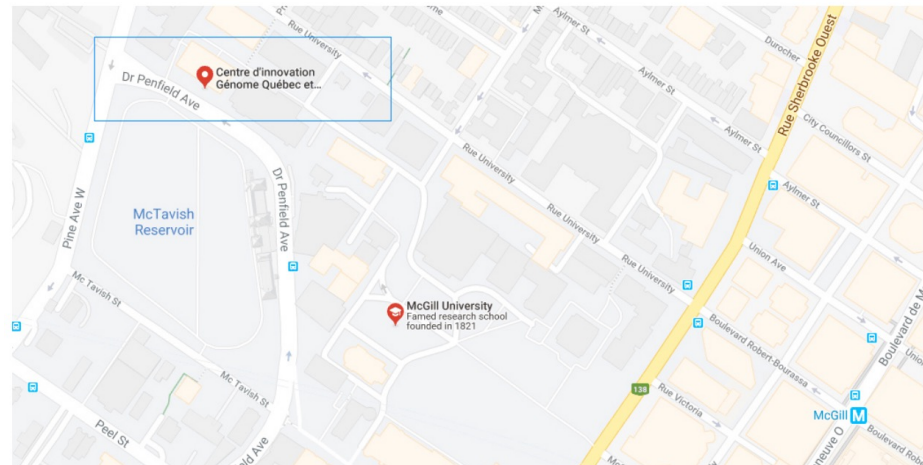
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Computational
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Contact



MiCoM McGill initiative in
Computational Medicine

McGill initiative in Computational Medicine
740, Dr. Penfield Avenue, Montreal, Quebec,
Canada, H3A 0G1
email: info-micm@mcgill.ca

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Outline

1. Task of differential gene expression: pairwise vs multigroup comparisons
2. Preprocess for fair comparison: Filtering / Normalization
3. Statistical testing: Exact test for pairwise
4. Linear regression
5. Poisson and negative binomial distributions
6. Generalized linear model
7. Sample analysis using edgeR (R notebook)

Differential Gene Expression (DGE) analysis

Learning objectives:

- Understand the goal of differential gene expression analysis
- Understand how experimental designs affect choice of analysis methods
- Know the general pipeline of DGE analysis

Differential Gene Expression Analysis: What?

- **Goal:** identify differentially expressed genes (DEGs)
 - **Determine how treatment and genotype affect expression by identifying significantly upregulated and downregulated genes**
 - **We can then analyze these identified genes to infer underlying biological mechanisms causing the variance**
- Common packages to perform DGE analysis:
 - **edgeR**
 - **DESeq2**
 - **limma**

Differential Gene Expression: How?

edgeR

- ❖ Empirical analysis of Digital Gene Expression in R.
- ❖ One of the *Bioconductor* packages

➤ “The mission of the Bioconductor project is to develop, support, and disseminate free open source software that facilitates rigorous and reproducible analysis of data from current and emerging biological assays.”

```
58  if (!requireNamespace("BiocManager", quietly = TRUE))
59    install.packages("BiocManager")
60
61  BiocManager::install("edgeR")
62
63  #load libraries ----
64  library(edgeR)
```

How to conduct DGE analysis with edgeR:

A - Pre-processing steps:

1. Convert genetic data into a DGE object → allows us to use analysis functions
2. Filter out low count genes → focusing on genes that are more likely to be involved in the biological processes of interest
3. Normalize data to enable comparison → consider each sample's library size and composition

B - Analysis steps:

1. Estimate the common and tag-wise dispersion
 - Common = overall variance
 - Tag-wise = gene specific variance
2. Perform exact test two sample groups of choice OR generalized linear model analysis

Preprocessing

Learning objectives:

- Understand why and how we filter counts
- Understand why and how we normalize counts
- Understand how PCA could help with quality control

Filtering Genes: Why?

- Remove by functional category
 - Ribosomal RNA
- Remove genes with very low counts: genes with very low counts across all libraries provide little evidence for differential expression.
 - Reduces computational cost
 - Potentially reduces noise
 - Risk losing novel genes

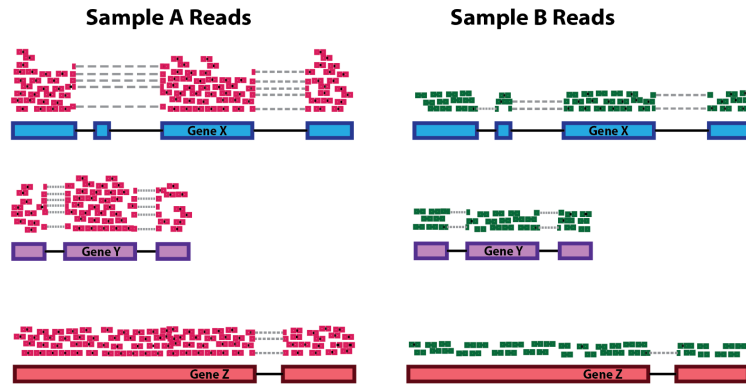
Filtering Genes: How?

- As a rule of thumb, genes are dropped if they can't possibly be expressed in **all the samples for any of the conditions**.
- Threshold setting:
 - Usually, a gene is required to have a count of 5-10 in a library to be considered expressed in that library.
 - Multiple thresholds should be tried to find the appropriate one: account for experimental design and library size

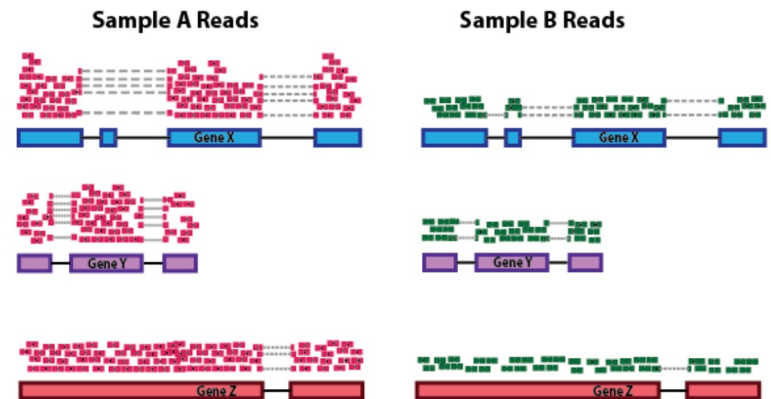
Use counts-per-million
which normalizes with
library size!!!

Normalization: Why?

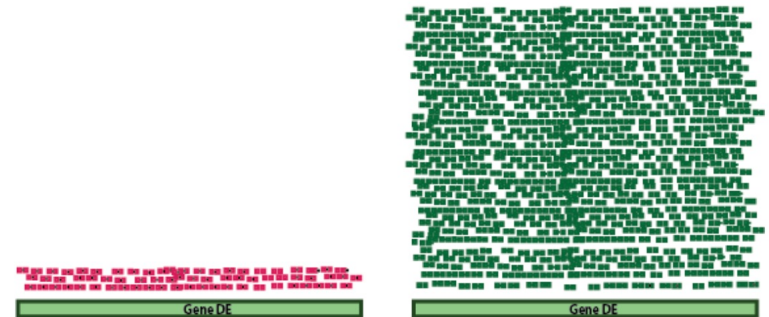
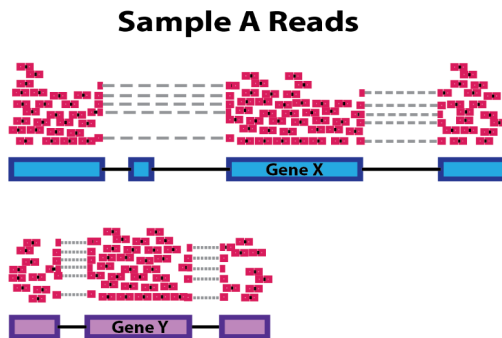
Sequencing Depth



RNA composition



Gene length



What are we assuming here by using CPM?
Same total expression

Normalization: How?

- Counts per million (CPM)
 - It normalizes RNA-seq data for sequencing depth but not gene length.
 - Raw counts are divided by the number of sequencing reads in your sample, multiplied by a million.
- Trimmed Mean of M-values (TMM)
 - “RNA-seq measures relative expression rather than absolute expression. This becomes important for differential expression analyses when a small number of genes are very highly expressed in some samples but not in others. If a small proportion of highly expressed genes consume a substantial proportion of the total library size for a particular sample, this will cause the remaining genes to be under-sampled for that sample. Unless this effect is adjusted for, the remaining genes may falsely appear to be down-regulated in that sample.”
 - Original paper:
<https://genomebiology.biomedcentral.com/articles/10.1186/gb-2010-11-3-r25>

TMM: How it works?

How TMM works:

- **Scaling factors:** The key idea is to compute a scaling factor for each sample that normalizes the counts. A reference sample (usually the median sample or a pseudo-reference) is chosen, and the counts of all other samples are compared to it.
- **M-values:** For each gene, an M-value is calculated, which is the log ratio of the expression levels between the sample and the reference:

$$M = \log_2 \left(\frac{\text{gene count in sample}}{\text{gene count in reference}} \right)$$

- The M-values for highly expressed genes (that dominate the library) or genes with very low counts (often noisy) can distort the normalization.
- To avoid this, TMM trims (excludes) genes that are too highly or too lowly expressed from the calculation of the scaling factor.
- **Final step:** After trimming, the mean of the remaining M-values is used to compute a scaling factor for each sample. These scaling factors are then used to normalize the raw counts, ensuring that differences in library size or sequencing depth do not skew the analysis.

From ChatGPT

Other normalization methods:

Gene length

Do we need this?
Gene length is the same for all samples.

- Reads per kilobase per million reads (RPKM):
 - RPKM is a within-sample normalization method which removes transcript-length and library size effects
 - First normalize by library size and then normalize by transcript length
 - Designed for single-end reads
- Fragments per kilobase per million mapped reads (FPKM):
 - FPKM is an extension of RPKM designed for paired-end reads
 - Each pair of reads are treated as one fragment here (if they both were mapped). This avoids counting a fragment twice which cannot be done by RPKM.
 - Renders the same output as RPKM on single-end reads.
- Transcripts per million (TPM)
 - TPM is an extension of RPKM where we first normalize by transcript length and then normalize by library size.
 - The sum of all TPMs is the same for each sample, making it easier to compare the proportion of reads mapped to a gene across samples.
 - Can also convert FPKM counts into TPM

When do we need this?
(1) Compare within sample
(2) Different conditions express different isoforms

Normalization Limitations

- Assume we have two identical samples
 - Knockout expression of Gene D in sample 2
- Fixed Library size means remaining counts are redistributed over the remaining genes

Gene	Sample 1	Sample 2
A	30	235
B	24	188
C	0	0
D	563	0
E	5	39
F	13	102
Total	635	635

In general, we don't expect the knocked-out genes to dominate the reads.

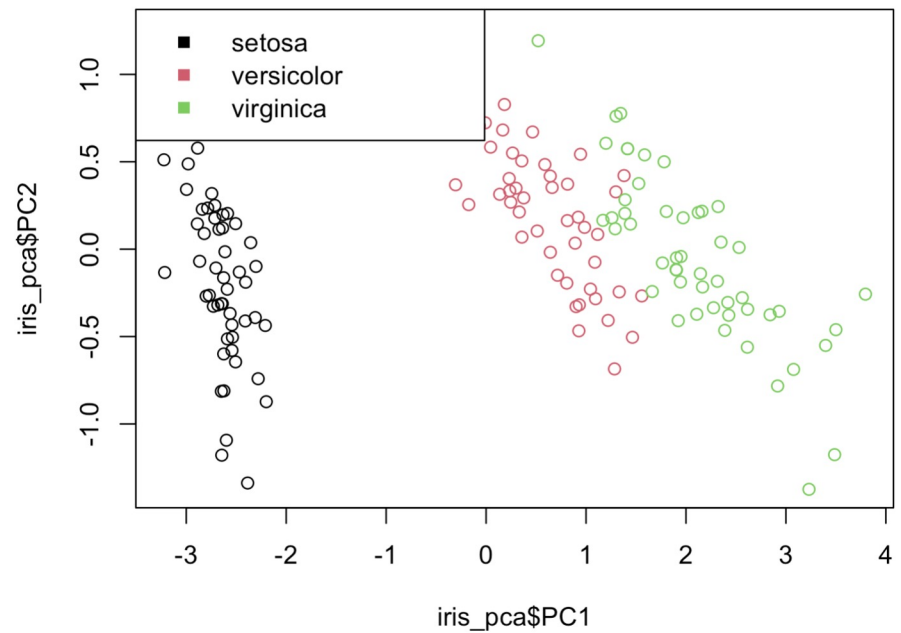
Dimensionality Reduction: Principal Component Analysis

What is dimensionality reduction?

The projection of high dimensional datasets into lower dimensional spaces, while retaining the meaningful properties of the data.

Key points:

- Keep most important features
 - Statistical significance
 - Correlated variables
- Create new features from original features via
 - Linear combinations (**PCA**)
 - Nonlinear combinations



Principal Component Analysis (PCA) on Gene Expression

Goal:

Determine the major contributing factors to gene expression variance between samples

How: Creates new variables called principal components (PC's) based on linear transformations of the original variables.

- PC's capture the variance in the dataset:
 - PC1 is the combination of variables with the highest variance,
 - PC2 is the combination of variables with the 2nd highest variance,
 - ...

Coding :

- PCA input variables = gene feature counts for each sample

Statistical Testing

Learning objectives:

- Understand the formulation and assumption of standard statistical tests
- Understand the interpretation of a p-value

Hypothesis Tests

Goal

- Determine if two sets of data are **different**
- Can approach this with **test statistics**
 - Is the difference **significant** or not?

Many Types of Tests

- Parametric tests – **make assumptions on the data distribution**
 - Z-score vs. t-test
 - ANOVA
- Non-parametric – **does not make assumptions on data dist.**
 - Permutation test

Hypothesis Formulation

Null Hypothesis

- Effect size is 0
- Difference between conditions is 0

Alternative Hypothesis

- Effect size is NOT 0
- Difference between conditions is NOT 0

We accept or reject the null based on the generated p-value (< 0.05)

Understanding Significance

Statistical significance **DOES NOT** imply causality

It is a measurement of the likelihood of the observed data happening by chance.

A p-value tells us the probability of seeing the observation **if the null hypothesis is true.**

Dispersions and the exact test

Exact test definition:

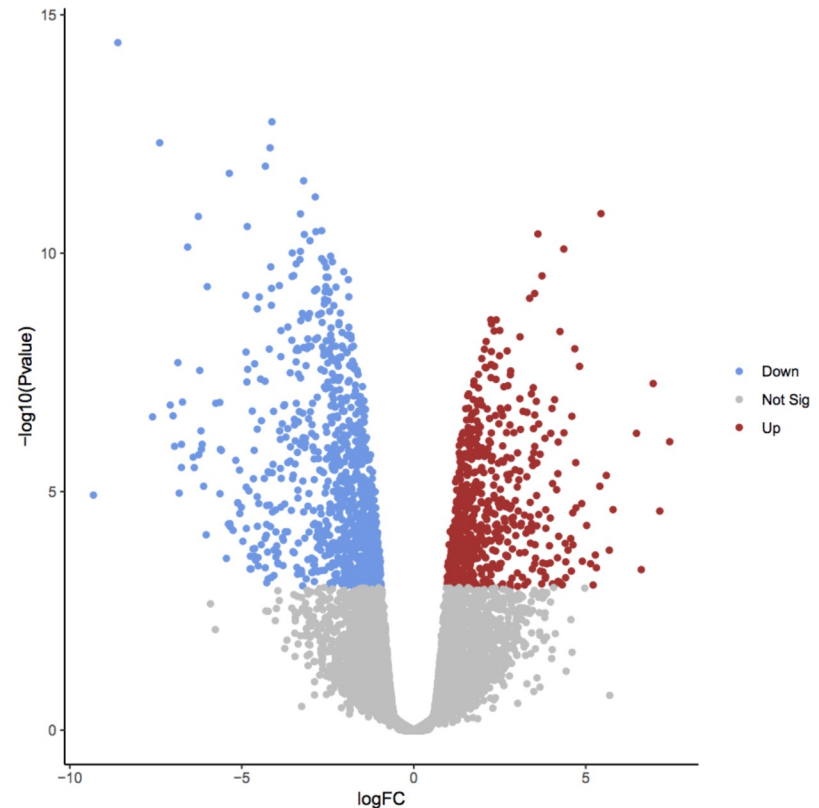
- Statistical method used to compare two groups (e.g., experimental conditions) when data **doesn't meet the assumptions** of traditional statistical tests (like the t-test).
- Identifies differentially expressed genes while accounting for the **overdispersion of RNA-seq data**

Dispersions to account for:

- **Common dispersion** = single value that represents the average level of variability across all genes
- **Tag-wise dispersion** = gene-specific variability based on its unique characteristics and the common dispersion

Volcano plot

- X axis: Fold change (log2)
- Y axis: pvalue/FDR ($-\log_{10}$)
- Goal: See what genes have a high FC AND high significance



What if we have multiple groups/dosages?

Example: test different siRNA knockdown conditions (25%, 50%, 100%) crossed with drug dosages (1 mM, 0.1 mM, 0.01 mM).

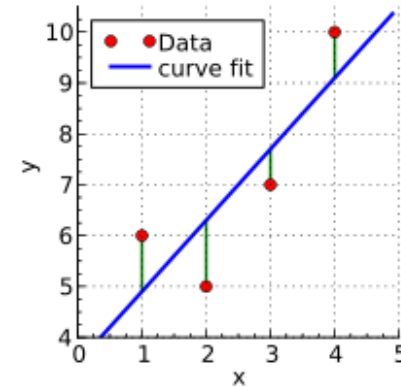
- 9 groups, how many pair-wise?
- **Solution: regression**

Linear Regression

Learning objectives:

- Understand the fundamental regression algorithms
- What is a coefficient/effect size

The Linear Model



Credit: Wikipedia

- The fundamental model for statistics & machine learning
- Follows a **Normal distribution**

Imagine a dataset with a dependent variable y and a set of descriptive features x

We want to learn what features in x are good predictors of y (what is their **effect size/coefficient** β)

$$y = \beta x + \varepsilon \quad \varepsilon \sim \text{Normal}(0, \sigma^2)$$

$$y \sim \text{Normal}(\beta x, \sigma^2)$$

Model Assumptions

1. Outcome is **continuous** and a **linear combination of predictors**
2. **Outcome is such that** $y_i \sim \text{Normal}(\beta x_i, \sigma^2)$
3. Predictors must not be perfectly correlated (linear combination)
4. For every observation i the error is:
 1. **Normally distributed**
 2. **Mean zero**
 3. **Homoskedastic** (same variance as other observations)
 4. **Independent** (not correlated)

Model Fitting

Probabilistic Approach

- **Maximum Likelihood Estimation** $\operatorname{argmax}_{\beta} \log(\text{Normal}(\beta x, \sigma^2))$

Machine Learning

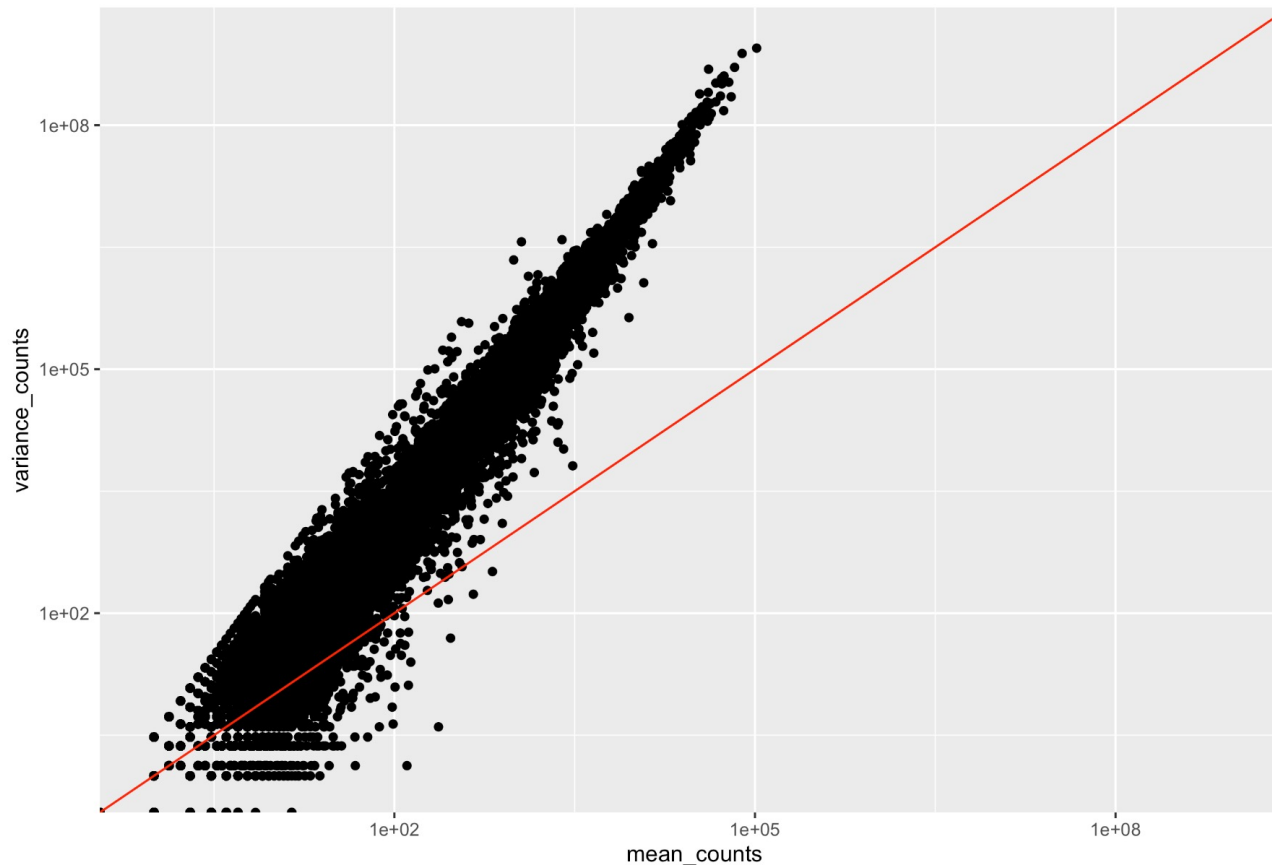
- **Mean Squared Error**

$$y = \beta x + \epsilon$$

$$\hat{y} = \hat{\beta} x$$

$$MSE = \frac{\sum_{i=1}^N (\hat{y}_i - y_i)^2}{N}$$

Are gene counts normally distributed? **NO**



Variance not
fixed

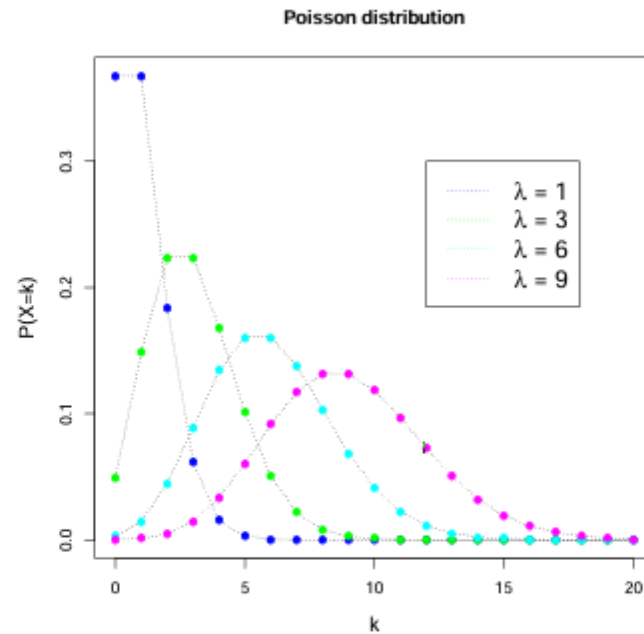
Adapted from: https://hbctraining.github.io/DGE_workshop_salmon_online/schedule/links-to-lessons.html

Poisson and Negative Binomial

Learning objectives:

- Understand properties of Poisson and negative binomial distributions
- Know why we use these to model RNA-seq count data

Poisson distribution

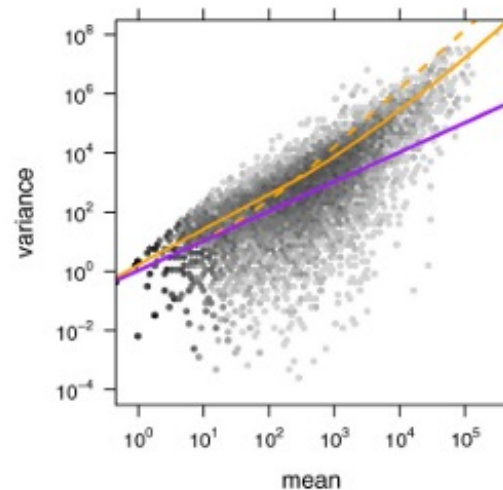


- For $X \sim \text{Poisson}(\lambda)$, both the mean and the variance are equal to λ

From Robinson

Negative binomial

Many studies have shown that the variance grows faster than the mean in RNAseq data. This is known as **overdispersion**.



- Mean count vs variance of RNA seq data. Orange line: the fitted observed curve. Purple: the variance implied by the Poisson distribution.

Anders S, Huber W (2010) Differential expression analysis for sequence count data. *Genome Biol* 11:R106.

From Robinson

What is Dispersion?

- A measure of spread or variability in the data

$$Var_{ij} = \mu_{ij} + \alpha_i \mu_{ij}^2$$

Rearranged:

$$\sqrt{\alpha_i} = \frac{Var_{ij} - \mu_{ij}}{\mu_{ij}}$$

Which is also the same as:

$$\sqrt{\alpha_i} = \frac{Var_{ij}}{\mu_{ij}} - 1$$

Adapted from: https://hbctraining.github.io/DGE_workshop_salmon_online/schedule/links-to-lessons.html

Generalized Linear Models

Learning objectives:

- Understand why and how to use generalized linear models to quantify differential gene expression

GLM Advantages

- Conditional distribution: allow Poisson or negative binomial in addition to Gaussian
- Link function $\rightarrow$ positive counts
- $E(Y_i | X_i) = e^{(X_i \cdot \beta)} > 0$

The linear model assumed a linear relationship between Y_i and X_i , since we assumed that $E(Y_i | X_i) = \mathbf{X}_i^T \beta$. In GLMs, we will allow a **link function** $g()$ that links the conditional mean to the covariates. Hence, in GLMs we have that $g(E(Y_i | X_i)) = \mathbf{X}_i^T \beta$. Note that each family has got a canonical link function, which is the identity link function $g(\mu) = \mu$ for Gaussian, the log link function $g(\mu) = \log \mu$ for Poisson, or the logit link function $g(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$ for Binomial.

Compare with log-transformed Linear Regression

- Linear model where Y_i is log-transformed: $E(\log Y_i)$, while in GLM, we have $\log E(Y_i)$
- This is not the same: in GLM, we are modelling a transformed version of the expected value, and we retransform to interpret the fit in terms of the mean of the response variables (gene count)

GLM for DEG

- Models Gene Expression using **Generalized Linear Models**
- Assumes counts are sampled from Negative-binomial distribution
- Use log as link function

Dispersion and Variability

The observed quadratic mean variance trend has motivated the use of the negative binomial distribution to model (bulk) RNA-seq gene expression data.

Anything
unexplained?

$$\begin{cases} Y_{gi} & \sim NB(\mu_{gi}, \phi_g) \\ \log \mu_{gi} & = \eta_{gi} \\ \eta_{gi} & = \mathbf{X}_i^T \beta_g + \log(O_{gi}) \end{cases}$$

coefficients β_g are log fold changes (with log link), tests on β_g tell us DE.

with

$$\text{var}[Y_{gi}] = \mu_{gi} + \phi_g \mu_{gi}^2$$

		Seq. technology		real expression
total variability	=	technical variability	+	biological variability
$\text{var}[Y_{gi}]$	=	μ_{gi}	+	$\phi_g \mu_{gi}^2$
total CV ²	=	$\frac{1}{\mu_{gi}}$	+	ϕ_g

From Berge and Clement

Normalization in GLM: Offset

$$\begin{cases} Y_{gi} & \sim N B(\mu_{gi}, \phi_g) \\ \log \mu_{gi} & = \eta_{gi} \\ \eta_{gi} & = \mathbf{X}_i^T \beta_g + \log(O_{gi}) \end{cases}$$

$$\mu_{ig} = \exp(\mathbf{X}_i^T \beta_g) \times O_{gi}$$

$$\log\left(\frac{\mu_{ig}}{O_{gi}}\right) = \mathbf{X}_i^T \beta_g$$

Coding Notebook in R

Theory: Dispersion estimation and Fitting GLM (references)

JOURNAL ARTICLE

Small-sample estimation of negative binomial dispersion, with applications to SAGE data

Mark D. Robinson ✉, Gordon K. Smyth

Biostatistics, Volume 9, Issue 2, April 2008, Pages 321–332,

<https://doi.org/10.1093/biostatistics/kxm030>

Published: 29 August 2007 **Article history** ▼

JOURNAL ARTICLE

Differential expression analysis of multifactor RNA-Seq experiments with respect to biological variation

Davis J. McCarthy, Yunshun Chen, Gordon K. Smyth ✉ [Author Notes](#)

Nucleic Acids Research, Volume 40, Issue 10, 1 May 2012, Pages 4288–4297,

<https://doi.org/10.1093/nar/gks042>

Published: 28 January 2012 **Article history** ▼

- Past MiCM slides: Intro to RNA-seq and Statistics in R (Adrien Osakwe)
- QLSC600 slides: myself and Megan Ng
- RNA-seq lecture by Peter N. Robinson
- Tutorial from Berge and Clement:
https://statomics.github.io/SGA/sequencing_countData.html
- Sample data from
Scheckel C, Drapeau E, Frias MA, Park CY et al. Regulatory consequences of neuronal ELAV-like protein binding to coding and non-coding RNAs in human brain. Elife 2016 Feb 19;5. PMID: 26894958



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D2R DNA
TO RNA

